

WHO WILL SPEAK FOR EARTH? POSSIBLE STRUCTURES FOR SHAPING A RESPONSE TO A SIGNAL DETECTED FROM AN EXTRATERRESTRIAL CIVILIZATION

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Abstract—I propose that the IAF and the International Astronomical Union establish a joint committee that will create a reply message to respond to a confirmed extraterrestrial signal. The goal of this committee will be to reach a consensus among the international intellectual community as to the contents of such a reply. If this reply message exists before a signal is detected, it will deflect pressure to send non-international replies and will focus attention on the desirability of having the Earth speak with one voice to our neighbor civilizations.

1. INTRODUCTION

The inauguration of the META system in 1985, and initiation of NASA's plans to build a broadband SETI detector system during the 1990s, represents a significant transition. SETI is no longer a completely speculative and occasional pursuit, but instead has become an ongoing activity, ripe for further development. As a result, the detection of an extraterrestrial civilization may well occur within the next ten or twenty years, if the number of radio-emitting civilizations in the Milky Way is of the order of 10^4 or larger.

Increases in SETI efforts have, quite understandably, not been matched by a corresponding attentiveness to the message(s) that might be broadcast from Earth once the reception of an extraterrestrial signal has been confirmed[1]. This might be taken to represent (i) a lack of confidence that SETI will succeed, (ii) a lack of interest in two-way communication, given the long travel time for messages, or (iii) a conservative approach to the issue, one that finds no need to determine the content of a reply until we have a confirmed signal. None of these seems logically supportable. In particular, I suggest that now is the time to begin consideration of what Earth will say in reply to an extraterrestrial intelligent signal.

2. RELEVANT FACTS

The great distances that separate us from neighboring civilizations imply long travel times for any messages, such as radio signals, sent at the speed of light. Estimated distances to the closest civilizations range from approximately 10^2 light-years (if millions of civilizations exist within the Milky Way) to 10^4 - 10^5 light-years (if only a few, or a few dozen, civilizations exist in our galaxy). In the present state of human consciousness, time intervals ranging from a few

hundred to hundreds of thousands of years may appear widely unacceptable, and interest in two-way communication relatively small. However, once a signal has been detected, it is likely that a tremendous human desire to respond to the signal will emerge, along with a significant, but minority, view holding that we would do well to maintain radio silence (which, however, has been contrafactual for the past forty years or more).

The same laws of nature that impose long travel times likewise assure that the first message leaving Earth will be the first to arrive at its destination. It is reasonable to expect that a civilization receiving a reply to its signal not only will detect and examine the first reply (if it is still in existence in a mode capable of doing so), but also will continue to look for further messages. It is likewise reasonable that if the sending civilization receives replies from various civilizations, it will send further information, or at least pay primary attention in doing so, to those civilizations that it judges to be "worthwhile." At a bare minimum, "worthwhile" probably means "not so disorganized as to send contradictory or jumbled messages." If this is true, there is a high price that a civilization such as ours will pay for not concentrating on a single, high-power message: we will go to the back of the line for a response to our reply.

Any signal detected on Earth will immediately reveal two key facts, the direction from which it arrives and the frequency at which it was broadcast. Furthermore, the strength of the arriving signal and its detailed characteristics will tell us much about the technological capabilities of the broadcasters, and the frequency dispersion caused by its passage through interstellar space will provide a good indication of the distance it has travelled. Thus even before the message content can be determined, we shall be in possession of facts sufficient to allow a reply to be sent with a high probability of detection (always assuming that

the sending civilization lasts long enough to keep listening for replies). The question is, how should Earth prepare and send any reply to a detected signal?

3. PRE-DETECTION PREPARATIONS OF A REPLY

Speculations on the topic of whether and how to reply to an extraterrestrial signal range (at least) from fears that fragments of terrestrial society might attempt to appropriate this function to hopes that the international community of scholars will function as planetary representatives. The first of these possibilities appears to founder on the facts of how human beings behave: I believe it is impossible that detection of an extraterrestrial signal could remain a secret for more than 24 hours. As to the second possibility, I submit that it is appropriate for the scientific community to take the lead in preparing a reply message, not because this will represent the only "acceptable" reply but because only scientists are prepared to look ahead - by an indefinite amount into the future - in an organized manner. If scientists wait until a signal is detected and then expect to be asked to create a reply, or then attempt to produce such a message, it seems likely that their voices will be lost in the uproar surrounding the detection.

I therefore propose that the International Astronautical Federation and the International Astronomical Union jointly create a committee that will encourage consultation to lead to a "worldwide" reply to other civilizations. This committee's mission would be to raise interest within the international scientific and scholarly community in crafting such a reply. The scientific community in turn may be able to awaken interest within the general public, but the key goal will be to lead toward a consensus - prior to the detection of an extraterrestrial message - as to what our reply will contain. Since it is too optimistic to assume that such a consensus could arise within the entire public, I confine my hopes to the suggestion that scientists take the lead and be known as having agreed on what Earth should say in reply to a detected signal.

Before any such detection, the construction of a reply message will be seen by large segments of both the scientific community and the public as a sterile pursuit, not worthy of serious attention. Nevertheless, the creation of a reply prior to the detection of a signal may prove crucial in allowing the scientific community to be heard more effectively in the debate that will

erupt as to what a reply should contain. The more widely the contents of a reply have been discussed and agreed upon, the more prevalent will be the attention given to those contents once detection occurs. Thus the fact that construction of a reply will be greeted with some derision is not all bad: it is as good a place as any to start the discussion, and only such discussion can lead to an increased awareness of the issues at stake upon detection of a signal. Following such detection, a decision may be taken - on a worldwide basis, it is profoundly to be hoped - to send a reply even before much of the message content has been deciphered. If so, the need to have an agreed-upon reply in existence seems larger than if agreement were reached not to send a reply until the message had been largely or completely understood.

4. POSSIBLE CONTENT OF A REPLY MESSAGE

Although it would be presumptuous to resolve in advance the output from the committee that I propose here, our technological capabilities illuminate certain aspects of possible reply messages. The Arecibo message, broadcast toward M13 in 1974, contained 1,679 bits of information, sent in the most straightforward manner as (effectively) on-off pulses in one of two frequency channels. However, to broadcast 10^7 - 10^8 bits of information requires only a few seconds, especially once the basic frequency (that of the arriving signal) has been established, so that nearby frequency channels are certain to be examined. This number of bits can contain, for example, the contents of all the sounds and visual images on the two Voyager records[2]. It is therefore apparent that any reply message will be able to contain a tremendous amount of information, quite possibly all that anyone might be able to suggest. This should make agreement as to the message content relatively easy, since nothing of crucial importance to any group need be totally omitted. (Arriving messages might well have developed from the same sociological principles.)

As Fasan has suggested[3], one basic rule that might be followed in constructing the reply message is that of truthfulness; that is, of not deliberately including any "information" known to be false. In view of the notorious difficulty of separating truth from falsity, even this simple rule may prove difficult to apply. Does exaggeration count as untruthfulness, as when Thomas suggested that we broadcast nothing but Bach[4]? Such issues deserve discussion for their scientific value (although they can always

be compromised by adding still more to the reply message), and still more because of the need, as I see it, to create a reply to have ready before the detection of a signal. Such a reply message, if it is the product of a representative segment of Earth's intellectual community, will help focus attention on the need for a planet-wide response, and will thereby help to avoid one society or another's insistence that it will speak for Earth.

REFERENCES

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